

Number Sense

Answer Key — Worked Solutions

MTH1W · Homework — Answer Key

Worked solutions. Method lines are shown for application and thinking items.

1. [L1] $(-12) + (+7) + (-3) = (-5) + (-3) = -8$.
2. [L1] Exponent: $2^2 = 4$. Multiply: $6 \times 3 = 18$.
 $40 - 18 + 4 = 26$.
3. [L1] Product rule (add exponents): $7^{3+2} = 7^5$.
4. [L1] Quotient rule (subtract exponents): $m^{5-2} = m^3$.
5. [L1] LCD = 24: $15/24 + 4/24 = 19/24$.
6. [L1] $0.24 = 24/100 = 6/25$; and $0.24 \times 100 = 24\%$.
7. [L1] $81 = 9^2$ is a perfect square, so $\sqrt{81} = 9$ (rational).
8. [L2] Left to right: $2/5 \times 3/4 = 6/20 = 3/10$.
 $\div 1/2 = 3/10 \times 2/1 = 6/10 = 3/5$.
9. [L2] Power of a power: $(3^2)^3 = 3^6$.
Quotient rule: $3^{6-4} = 3^2 = 9$.
10. [L2] Markup = $0.25 \times 60 = \$15$.
New price = $60 + 15 = \$75$ (or $60 \times 1.25 = \$75$).
11. [L2] As decimals: $-1/3 \approx -0.333$, $1/5 = 0.2$.
Compare: $-0.5 < -0.4 < -0.333 < 0.2$.
Answer: -0.5 , -0.4 , $-1/3$, $1/5$.
12. [L2] Set up the proportion $1 \text{ cm} : 25 \text{ km} = 7 \text{ cm} : d$.
 $d = 7 \times 25 = 175 \text{ km}$.
13. [L2] Unit rate: $8.40 \div 3 = \$2.80/\text{kg}$.
5 kg: $2.80 \times 5 = \$14.00$.
14. [L2] $5^2 = 25$ and $6^2 = 36$, and $25 < 30 < 36$.
So $\sqrt{30}$ lies between 5 and 6.
15. [L3] The sale price is 80% of the original: $0.80 \times p = 96$.
 $p = 96 \div 0.80 = \$120$. Check: $120 \times 0.80 = \$96 \checkmark$.
16. [L3] Error: for a power of a power you MULTIPLY the exponents, not add them.
 $(2^3)^2 = 2^{3 \times 2} = 2^6 = 64$, not $2^5 = 32$.

- 17. [L3]** Juice now = $0.40 \times 500 = 200$ mL. Let x = mL of pure juice added.
 $(200 + x) / (500 + x) = 0.50 \rightarrow 200 + x = 250 + 0.5x$.
 $0.5x = 50 \rightarrow x = 100$ mL of pure juice.
- 18. [L3]** Times: $36 \div 12 = 3$ h and $20 \div 10 = 2$ h, so riding time = 5 h.
Average speed = total distance $\div$ riding time = $56 \div 5 = 11.2$ km/h.